ARCHITECTURE DECISION RECORDS · 14 DECISIONS

ADR Cards

One card per decision. Each card shows the number, scope, context, status, the decision in one breath, and why it was taken. Generated from the ADR front matter; the full record is the linked file under `obsidian/V3-Platform/Architecture/ADR/`.

ADR	SCOPE · CONTEXT	TITLE	STATUS	DATE
ADR-046	LAB ORGANIZATIONS	Transporter Aggregate Boundary: Shared Identity, Separate Vetting	Accepted	2026-08-20
ADR-047	LAB ORGANIZATIONS	Temporal Saga Design for TransporterProfile Vetting	Accepted	2026-08-20
ADR-048	LAB ORGANIZATIONS	Document Blob Storage: NATS Object Store	Accepted	2026-08-20
ADR-049	LAB ORGANIZATIONS	Cross-Aggregate Concurrency: Two Mechanisms, One Guard	Accepted	2026-08-20
ADR-050	LAB ORGANIZATIONS	GIT Certificate Change Log Needs a Provenance Source	Accepted	2026-08-22
ADR-051	LAB ORGANIZATIONS	Entity Identity Is a ULID, Not a UUID	Accepted	2026-08-24
ADR-052	LAB DATA	One Postgres Instance, One Database and One Role per Service	Accepted	2026-09-03
ADR-053	V3 DATA	Use Shared PostgreSQL Instances by Default	Accepted	2026-09-03
ADR-054	V3 PLATFORM	Portability Rules for Multi-Region Deployment and Integration	Accepted	2026-09-07
ADR-055	LAB PLATFORM	Split the Lab Compose File into Cell Tiers, and Make Every Host Port a Variable	Accepted	2026-09-08
ADR-056	V3 PLATFORM	Five Independent Axes — Region, Tenant, Account, Cell, JetStream Domain	Accepted	2026-09-18
ADR-057	LAB NATS	An Auth Callout in Operator Mode Turns a WorkOS Identity into NATS Access	Proposed	2026-10-09
ADR-058	V3 OBSERVABILITY	Trace Every Service with the OpenTelemetry SDK, Not Over NATS	Proposed	2026-10-18
ADR-059	LAB OBSERVABILITY	OpenTelemetry Replaces natstrace; Spans Go to a Collector and Tempo, Not Over NATS	Accepted	2026-10-16

- lab — Tech Lab / Dictionary POC decision, implemented in this repo
- v3 — Proposed Linebooker V3 platform principle
- Accepted
- Proposed

046

LAB · ORGANIZATIONS

Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ACCEPTED 2026-08-20 ADR-046-lab-organizations-transporter-aggregate-split.md

DECISION

Organization stays the single identity aggregate for Shipper and Transporter. Vetting state lives in a new event-sourced TransporterProfile aggregate.

WHY

Identity is current-state CRUD. Vetting has a lifecycle that must replay and audit. One aggregate would force both onto one persistence model.

RELATIONSHIPS

related ADR-047, ADR-048, ADR-049, ADR-050, ADR-051

LINKED DECISIONS

ADR-047 Temporal Saga Design for TransporterProfile Vetting

ADR-048 Document Blob Storage: NATS Object Store

ADR-049 Cross-Aggregate Concurrency: Two Mechanisms, One Guard

ADR-050 GIT Certificate Change Log Needs a Provenance Source

ADR-051 Entity Identity Is a ULID, Not a UUID

047

LAB · ORGANIZATIONS

Temporal Saga Design for TransporterProfile Vetting

ACCEPTED 2026-08-20 ADR-047-lab-organizations-transporter-vetting-temporal-saga.md

DECISION Vetting runs as a Temporal workflow with a two-branch parallel saga. Activities publish the domain events. Compensation is forward-only.

WHY A multi-day, human-in-the-loop approval with real compensation needs durable timers, signals and saga support. That is what Temporal is for.

RELATIONSHIPS

related ADR-046, ADR-048, ADR-049, ADR-050

LINKED DECISIONS

ADR-046 Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-048 Document Blob Storage: NATS Object Store

ADR-049 Cross-Aggregate Concurrency: Two Mechanisms, One Guard

ADR-050 GIT Certificate Change Log Needs a Provenance Source

048

LAB · ORGANIZATIONS

Document Blob Storage: NATS Object Store

ACCEPTED 2026-08-20 ADR-048-lab-organizations-document-storage-nats-object-store.md

DECISION

Compliance document bytes go in a NATS Object Store bucket per tenant account, not S3, MinIO or GCS.

WHY

The lab evaluates NATS patterns and tenant isolation already comes from the NATS account. The cost is real limits on size and durability, and the ADR records them.

RELATIONSHIPS

related ADR-046, ADR-047, ADR-050

LINKED DECISIONS

ADR-046 Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-047 Temporal Saga Design for TransporterProfile Vetting

ADR-050 GIT Certificate Change Log Needs a Provenance Source

049

LAB · ORGANIZATIONS

Cross-Aggregate Concurrency: Two Mechanisms, One Guard

ACCEPTED 2026-08-20 ADR-049-lab-organizations-cross-aggregate-concurrency.md

DECISION

Organization uses optimistic row versioning. TransporterProfile uses per-subject expected-sequence on the TRANSPORTER stream. One guard sits across both.

WHY

The two aggregates persist differently, so no single mechanism covers both. The cross-aggregate invariant is a one-time gate, not a running constraint.

RELATIONSHIPS

related ADR-046, ADR-047, ADR-051

LINKED DECISIONS

ADR-046 Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-047 Temporal Saga Design for TransporterProfile Vetting

ADR-051 Entity Identity Is a ULID, Not a UUID

050

LAB · ORGANIZATIONS

GIT Certificate Change Log Needs a Provenance Source

ACCEPTED 2026-08-22

ADR-050-lab-organizations-git-certificate-change-log-provenance.md

DECISION

The GOODS_IN_TRANSIT document is event-sourced onto the TRANSPORTER stream. Its Postgres row becomes a projection. The other four document types stay CRUD.

WHY

A change log needs an event history that did not exist. Only the GIT certificate has a business need for one, so only it pays the event-sourcing cost.

RELATIONSHIPS

related ADR-046, ADR-047, ADR-048

LINKED DECISIONS

ADR-046 Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-047 Temporal Saga Design for TransporterProfile Vetting

ADR-048 Document Blob Storage: NATS Object Store

051

LAB · ORGANIZATIONS

Entity Identity Is a ULID, Not a UUID

ACCEPTED 2026-08-24 ADR-051-lab-organizations-ulid-entity-identity.md

DECISION organizations-service mints 26-character ULIDs in Go. ID columns are TEXT with no default. shipping-service and accounts-service stay on UUID.

WHY IDs sit in NATS subject tokens and KV keys, so they must be subject-safe and immutable. ULIDs sort by time and never contain a dot.

RELATIONSHIPS

related ADR-046, ADR-049

LINKED DECISIONS

ADR-046 Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-049 Cross-Aggregate Concurrency: Two Mechanisms, One Guard

052

LAB · DATA

One Postgres Instance, One Database and One Role per Service

ACCEPTED 2026-09-03 ADR-052-lab-data-one-postgres-instance-database-per-service.md

DECISION

The six lab services share one Postgres container. Each owns one database and one role. CONNECT is revoked from PUBLIC and granted to the owner only.

WHY

Seven containers bought process isolation nothing needed. Per-role CONNECT privileges keep the real guarantee: a service cannot read another service's tables.

RELATIONSHIPS

applies ADR-053

LINKED DECISIONS

ADR-053 Use Shared PostgreSQL Instances by Default

053

V3 · DATA

Use Shared PostgreSQL Instances by Default

ACCEPTED 2026-09-03 ADR-053-v3-data-shared-postgres-by-default.md

DECISION

Share PostgreSQL infrastructure by default within a region. Give every service its own logically isolated database, credentials and migrations. Move a database to dedicated infrastructure only on a demonstrated requirement.

WHY

Dedicated instances per database cost more, fragment capacity and multiply operational work. Logical isolation by database, role and privileges keeps service ownership strong without that cost.

RELATIONSHIPS

applied by ADR-052

LINKED DECISIONS

ADR-052 One Postgres Instance, One Database and One Role per Service

054

V3 · PLATFORM

Portability Rules for Multi-Region Deployment and Integration

ACCEPTED 2026-09-07 ADR-054-v3-platform-portability-rules-multi-region.md

DECISION

Every managed or vendor-specific dependency sits behind a port with at least two working adapters, a region runs on plain compute, disk and network, and no global component holds a copy of regional data. Ten numbered rules govern multi-region deployment, clustering and integration, including which of three cross-border shapes a residency or sovereignty requirement selects.

WHY

Linebooker V3 must be liftable between clouds and into a customer's own datacentre. A lock-in decision is cheap to take and expensive to reverse, so the constraint has to be written down before the choices are made, not after.

RELATIONSHIPS

related ADR-048, ADR-053

LINKED DECISIONS

ADR-048 Document Blob Storage: NATS Object Store

ADR-053 Use Shared PostgreSQL Instances by Default

055

LAB · PLATFORM

Split the Lab Compose File into Cell Tiers, and Make Every Host Port a Variable

ACCEPTED 2026-09-08 ADR-055-lab-platform-compose-split-cell-and-global.md

DECISION

The one flat docker-compose.yml becomes three tier files under deploy/cell/ (infra, runtime, dedicated) plus an include-only compose.yaml, and one file under deploy/global/ (control). Every published host port is a ${VAR:-default} fed by one env file per cell, and container_name is removed everywhere.

WHY

A second local region could not start. Every host port and every container name was a literal, so two cells collided on both. Files split by when a thing ships; projects split by where it runs.

RELATIONSHIPS

applies ADR-054

related ADR-052

LINKED DECISIONS

ADR-052 One Postgres Instance, One Database and One Role per Service

ADR-054 Portability Rules for Multi-Region Deployment and Integration

056

V3 · PLATFORM

Five Independent Axes — Region, Tenant, Account, Cell, JetStream Domain

ACCEPTED 2026-09-18 ADR-056-v3-platform-five-axes-placement-ownership-authority.md

DECISION

Region, tenant, account, cell and JetStream domain are five independent axes, each answering a different question, and no two of them may be defined as permanently equal. The bare word "domain" is reserved for the NATS JetStream sense; the DNS sense is renamed "market host" and the DDD sense is always written "business domain".

WHY

A team review collapsed placement, ownership, authority, deployment and storage into one nested hierarchy. Collapsing any two of them removes the ability to move one without the other, and the word "domain" already carries three unrelated meanings in the same documents.

RELATIONSHIPS

related ADR-054

LINKED DECISIONS

ADR-054 Portability Rules for Multi-Region Deployment and Integration

057

LAB · NATS

An Auth Callout in Operator Mode Turns a WorkOS Identity into NATS Access

PROPOSED 2026-10-09 ADR-057-lab-nats-external-identity-auth-callout.md

DECISION

A Go auth callout service, registered on a dedicated AUTH account in operator mode, validates the WorkOS access token on every new connection and returns a signed NATS user for the mapped account (ORDERS) and role permissions. Proposed until demo 07's exercises validate it.

WHY

A person's identity and role live in WorkOS, which NATS cannot read. Auth callout is the agreed direction because it decides at connect time from the live token, stores no NATS user, and keeps operator mode as the trust base.

RELATIONSHIPS

none recorded

LINKED DECISIONS

none

058

V3 · OBSERVABILITY

Trace Every Service with the OpenTelemetry SDK, Not Over NATS

PROPOSED 2026-10-18 ADR-058-v3-observability-opentelemetry-tracing.md

DECISION

Every V3 service traces with the standard OpenTelemetry SDK through one shared module per language, and spans leave over OTLP to a disk-buffered agent, never over NATS accounts. An outside call always starts a new trace, and a span's tenant comes only from trusted context.

WHY

The lab's private tracer had two defects in code the platform would have to own. The standard SDK already does creation, propagation, sampling and export, so only the NATS and tenant rules need platform code.

RELATIONSHIPS

applied by ADR-059

related ADR-055

LINKED DECISIONS

ADR-059 OpenTelemetry Replaces natstrace; Spans Go to a Collector and Tempo, Not Over NATS

ADR-055 Split the Lab Compose File into Cell Tiers, and Make Every Host Port a Variable

059

LAB · OBSERVABILITY

OpenTelemetry Replaces natstrace; Spans Go to a Collector and Tempo, Not Over NATS

ACCEPTED 2026-10-16 ADR-059-lab-observability-opentelemetry-replaces-natstrace.md

DECISION

The lab deletes natstrace and the NATS trace path; six services trace through shared/otelnats and export OTLP/HTTP to an OTel Collector with a disk queue, which writes to Tempo. Operators read traces through observability-service over NATS api.* or in Grafana; tenants read nothing.

WHY

The Phase 28 bridge had two defects (tenant spans never exported; outbound spans named the callee), and fixing them meant keeping a private tracer. The SDK made the lab's own code small enough to fit one package.

RELATIONSHIPS

applies ADR-058

related ADR-055

LINKED DECISIONS

ADR-058 Trace Every Service with the OpenTelemetry SDK, Not Over NATS

ADR-055 Split the Lab Compose File into Cell Tiers, and Make Every Host Port a Variable